



The Printer Working Group

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IPP Firmware Update Extensions v1.0 (FWUPDATE)

Status: Prototype

Abstract: This specification defines new IPP attributes, values, and operations to enable a Client to request a Printer to check for new Firmware, report information about discovered new Firmware, and to install the available new Firmware.

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<https://ftp.pwg.org/pub/pwg/general/process/pwg-process-4.pdf>

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To meet this objective, the PWG documents the results of their work as open standards that define print related protocols, interfaces, procedures, and conventions. A PWG standard is a stable, well understood, and technically competent specification that is widely used with multiple independent and interoperable implementations. Printer manufacturers and vendors of printer related software benefit from the interoperability provided by voluntary conformance to these standards.

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Table of Contents

91		
92	1. Introduction.....	7
93	2. Terminology.....	7
94	2.1 Conformance Terminology.....	7
95	2.2 Printing Terminology	7
96	2.3 Protocol Role Terminology.....	8
97	2.4 Other Terminology.....	8
98	2.5 Acronyms and Organizations	9
99	3. Requirements	9
100	3.1 Rationale.....	9
101	3.2 Use Cases	10
102	3.2.1 Check for New Firmware	10
103	3.2.2 Install New Firmware Immediately	10
104	3.2.3 Install New Firmware Later.....	10
105	3.3 Exceptions	10
106	3.3.1 Update Firmware Successful	10
107	3.3.2 Update Firmware Unsuccessful	10
108	3.4 Out of Scope.....	11
109	3.5 Design Requirements.....	11
110	4. Model.....	12
111	4.1 Firmware Update Interaction Types.....	12
112	4.1.1 Directed Firmware Update	13
113	4.1.2 Autonomous Firmware Update	13
114	4.1.3 Triggered Autonomous Firmware Update.....	14
115	4.2 New Firmware Available Status.....	15
116	4.3 Requesting Printer to Check for New Firmware	16
117	4.4 Requesting Printer to Update Firmware	16
118	4.5 Selecting a Printer Firmware Repository	17
119	4.6 Updating the Printer Firmware Update Feature Policy	17
120	5. New Operations.....	18
121	5.1 Check-For-New-Printer-Firmware.....	18
122	5.1.1 Check-For-New-Printer-Firmware Request	18
123	5.1.2 Check-For-New-Printer-Firmware Response	18
124	5.2 Update-Printer-Firmware.....	19
125	5.2.1 Update-Printer-Firmware Request	20
126	5.2.2 Update-Printer-Firmware Response	21
127	6. New Attributes	22
128	6.1 Operation Attributes	22
129	6.1.1 delay-update-until (type2 keyword).....	22
130	6.1.2 delay-update-until-date-time (dateTime).....	22
131	6.2 Printer Description Attributes	22
132	6.2.1 delay-update-until-supported (1setOf type2 keyword)	22
133	6.2.2 printer-new-firmware-policy-configured (type2 keyword)	22
134	6.2.3 printer-new-firmware-policy-supported (1setOf type2 keyword)	23
135	6.2.4 printer-firmware-repository-uri-configured (uri).....	23
136	6.2.5 printer-firmware-repository-uri-supported (1setOf uri).....	23

137	6.3 Printer Status Attributes	23
138	6.3.1 printer-firmware-info-uri (uri).....	23
139	6.3.2 printer-new-firmware-check-date-time (dateTime no-value unknown)	24
140	6.3.3 printer-new-firmware-delayed-until (type2 keyword)	24
141	6.3.4 printer-new-firmware-delayed-until-date-time (dateTime)	24
142	6.3.5 printer-new-firmware-info-uri (uri no-value unknown).....	24
143	6.3.6 printer-new-firmware-k-octets (1setOf integer(1:MAX) no-value unknown) ...	24
144	6.3.7 printer-new-firmware-name (1setOf name(MAX) no-value unknown)	25
145	6.3.8 printer-new-firmware-patches (1setOf text(MAX) no-value unknown).....	25
146	6.3.9 printer-new-firmware-string-version (1setOf text(MAX) no-value unknown) ..	25
147	6.3.10 printer-new-firmware-urgency (type2 keyword no-value unknown).....	25
148	6.3.11 printer-new-firmware-version (1setOf octetString(MAX) no-value unknown)	25
149	7. New Values for Existing Attributes	26
150	7.1 ipp-features-supported	26
151	7.2 operations-supported (1setOf type2 enum).....	26
152	7.3 printer-state-reasons (1setOf type2 keyword)	26
153	7.3.1 firmware-repository-access-error	26
154	7.3.2 firmware-repository-unreachable	26
155	7.3.3 new-firmware-installation-delay-until-specified	27
156	7.3.4 new-firmware-available.....	27
157	7.3.5 new-firmware-acquisition-failure	27
158	7.3.6 new-firmware-acquisition-in-progress	27
159	7.3.7 new-firmware-acquisition-success	27
160	7.3.8 new-firmware-activation-failure	27
161	7.3.9 new-firmware-activation-in-progress	27
162	7.3.10 new-firmware-activation-success	27
163	7.3.11 new-firmware-cleanup-failure	27
164	7.3.12 new-firmware-cleanup-in-progress	28
165	7.3.13 new-firmware-cleanup-success.....	28
166	7.3.14 new-firmware-installation-failure	28
167	7.3.15 new-firmware-installation-in-progress	28
168	7.3.16 new-firmware-installation-success	28
169	7.3.17 new-firmware-recovery-failure.....	28
170	7.3.18 new-firmware-recovery-in-progress	28
171	7.3.19 new-firmware-recovery-success	28
172	7.3.20 new-firmware-validation-failure	28
173	7.3.21 new-firmware-validation-in-progress.....	29
174	7.3.22 new-firmware-validation-success	29
175	8. Conformance Requirements	29
176	8.1 Printer Conformance Requirements	29
177	8.2 Client Conformance Requirements.....	29
178	9. Internationalization Considerations	29
179	10. Security and Privacy Considerations.....	30
180	11. Implementation and Deployment Considerations	31
181	12. IANA Considerations.....	32
182	12.1 Attribute Registrations	32

183	12.2 Type2 keyword Registrations.....	33
184	12.3 Type2 enum Registrations	34
185	12.4 Operation Registrations	34
186	13. References	35
187	13.1 Normative References.....	35
188	13.2 Informative References	37
189	14. Authors	37
190	15. Change History.....	38
191	15.1 July 29, 2026	38
192	15.2 July 16, 2026	38
193	15.3 May 6, 2026.....	39
194	15.4 April 22, 2026	39
195	15.5 February 25, 2026	39
196	15.6 January 27, 2026.....	40
197	15.7 November 14, 2025.....	40
198	15.8 October 29, 2025.....	41
199	15.9 May 6, 2025.....	41
200	15.10 October 24, 2024.....	41
201	15.11 October 10, 2024.....	41
202	15.12 September 11, 2024.....	41
203	15.13 August 26, 2024	42

List of Figures

207	Figure 1 - Firmware Update Phases.....	12
208	Figure 2 - Firmware Update Conceptual Model	13
209	Figure 3 - Triggered Autonomous Firmware Update Sequence Diagram.....	15

List of Tables

213	Table 1 - New Firmware Status Attributes.....	16
214	Table 2 – Keywords for the "delay-update-until" attribute.....	22
215	Table 3 – Keywords for "printer-new-firmware-policy-supported"	23
216	Table 4 – Keywords for "printer-new-firmware-urgency"	25
217	Table 5 – Enum values for "operations-supported"	26

1. Introduction

Printer vendors provide new Firmware for Printers, and Printers support various mechanisms to install new Firmware. IPP System Service v1.1 [PWG 5100.22] provides a way to push new Firmware through the System. Conversely, some Printers are designed and implemented to contact a Firmware Repository on their own, and to update themselves with available new Firmware.

This specification defines new IPP attributes, values, and operations to enable a Client to request a Printer to determine whether there is available new Firmware, report information about available new Firmware, and request the Printer to install available new Firmware.

2. Terminology

2.1 Conformance Terminology

Capitalized terms, such as MUST, MUST NOT, RECOMMENDED, REQUIRED, SHOULD, SHOULD NOT, MAY, and OPTIONAL, have special meaning relating to conformance as defined in Key words for use in RFCs to Indicate Requirement Levels [BCP14]. This specification defines the following additional capitalized conformance terms:

CONDITIONALLY REQUIRED: A MUST conformance requirement that applies only when a specified condition is true.

DEPRECATED: A SHOULD NOT conformance requirement for previously defined and approved protocol elements that are planned to be removed from use.

OBSOLETE: A MUST NOT conformance requirement for previously defined and approved protocol elements that have been removed from use.

2.2 Printing Terminology

The following printing terms are used in this document:

Administrator: An End User who is also authorized to manage all aspects of a Printer, including creating the Printer instances and controlling the authorization of other End Users and Operators [STD92].

Authenticated User: The same as "authenticated user" defined in [STD92] RFC 8011 section 9.3.

End User: An End User is a person or software process that is authorized to perform basic printing functions, including finding/locating a Printer, creating a local instance of a Printer,

250 viewing Printer status, viewing Printer capabilities, submitting a Print Job, viewing Print Job
251 status, and altering the attributes of a Print Job [STD92].

252 *Logical Device*: a print server, software service, or gateway that processes jobs and either
253 forwards or stores the processed Job or uses one or more Physical Devices to render
254 output [STD92].

255 *Operator*: An Operator is an End User that also has special rights on the Printer. The
256 Operator typically monitors the status of the Printer and also manages and controls the
257 Jobs at the Printer [STD92]. The Operator is allowed to query and control the Printer, Jobs,
258 and Documents based on site policy.

259 *Printer*: Listener for incoming IPP session requests and receiver of incoming IPP operation
260 requests (HTTP/1.1 server, as defined in [RFC7230]) that represents one or more Physical
261 Devices or a Logical Device. [STD92].

262 *Physical Device*: a hardware implementation of an endpoint device, e.g., a marking engine,
263 a fax modem, etc. [STD92]

264 **2.3 Protocol Role Terminology**

265 The following protocol roles are defined to specify unambiguous conformance
266 requirements:

267 *Client*: Initiator of outgoing connections and sender of outgoing operation requests
268 (Hypertext Transfer Protocol -- HTTP/1.1 [STD99] User Agent) [STD92].

269 *Printer*: Listener for incoming connections and receiver of incoming operation requests
270 (Hypertext Transfer Protocol -- HTTP/1.1 [STD99] Server) that represents one or more
271 Physical Devices or a Logical Device [STD92].

272 **2.4 Other Terminology**

273 This specification defines or adopts the following terms:

274 *Acquisition*: The new Firmware installation phase when the available new Firmware is
275 downloaded from the Firmware Repository to the Printer.

276 *Activation*: The new Firmware installation phase when the installed new Firmware is
277 activated for use.

278 *Cleanup*: The new Firmware installation phase when new firmware installation content is
279 no longer needed and is deleted from the Printer.

280 *Discovery*: The new Firmware installation phase when available new Firmware is found by
281 the Printer in the configured Firmware Repository.

282 *Installation*: The new Firmware installation phase when the acquired new Firmware is
283 installed to make it available for Activation.

284 *Recovery*: The new Firmware installation phase when the existing or previously installed
285 Firmware is restored for use following an error with one of the other phases (Acquisition,
286 Validation, Installation, Activation).

287 *Validation*: The new Firmware installation phase when the integrity of downloaded new
288 Firmware is confirmed by the Printer.

289 *Firmware*: Persistent computer instructions and data embedded in the Printer that provides
290 the basic functions of that device. Firmware is only replaced during a specialized update
291 process. [PWG 5110.1]

292 *Firmware Repository*: A service that hosts available Firmware packages.

293 *Management Device*: A device or set of devices that is used to manage Printers.

294 **2.5 Acronyms and Organizations**

295 This specification defines the following acronyms and organizations:

296 *IANA*: Internet Assigned Numbers Authority, <https://www.iana.org/>

297 *IETF*: Internet Engineering Task Force, <https://www.ietf.org/>

298 *ISO*: International Organization for Standardization, <https://www.iso.org/>

299 *PWG*: Printer Working Group, <https://www.pwg.org/>

300 **3. Requirements**

301 **3.1 Rationale**

302 Given the following existing specifications:

- 303 1. The IPP Driver Replacement Extensions v2.0 [PWG 5100.13]; and
- 304 2. IPP System Service v1.1 [PWG 5100.22]

305 And given the existence of a wider range of Firmware update use cases as described in
306 section 3.2, this IPP Firmware Update Extensions v1.0 specification should:

- 307 1. Define new attributes that will supply the versions of available new Firmware the
308 Printer is able to detect by querying its Firmware Repository;
- 309 2. Define new values for indicating the Printer's state while it is installing new
310 Firmware;

- 311 3. Define new operations to cause the Printer to check for new Firmware and install
312 new Firmware; and
313 4. Register all of these with IANA.

314 **3.2 Use Cases**

315 **3.2.1 Check for New Firmware**

316 Rafael has a document on his phone and chooses to print it. While he is looking at his print
317 options, his phone indicates that new firmware is available for his printer. It offers to remind
318 him later. He approves of being reminded later and prints his document.

319 **3.2.2 Install New Firmware Immediately**

320 Rafael is using his phone when it presents him with a notification informing him that new
321 firmware is available for his printer, to see if he wishes to approve the update to occur. It
322 gives him the option to install the new firmware immediately, install the new firmware at a
323 specified later time, or not install the new firmware. Rafael chooses to install the new
324 firmware immediately. The printer installs the new firmware immediately.

325 **3.2.3 Install New Firmware Later**

326 Rafael is using his phone when it presents him with a notification informing him that new
327 firmware is available for his printer, to see if he wishes to approve the update to occur. It
328 gives him the option to install the new firmware immediately, install the new firmware at a
329 specified later time, or not install the new firmware. Rafael chooses to install the new
330 firmware at 11:30pm that evening, then gets back to his email. He goes to sleep at
331 10:30pm.

332 At 11:30pm that evening, the printer install the new firmware as configured earlier.

333 **3.3 Exceptions**

334 The following subsections define exceptions in addition to those defined in the Internet
335 Printing Protocol/1.1 [STD92].

336 **3.3.1 Update Firmware Successful**

337 Rafael wakes up in the morning, and his phone presents a notification that his printer
338 attempted to install new Firmware at the chosen time, and that the new Firmware was
339 installed successfully.

340 **3.3.2 Update Firmware Unsuccessful**

341 Rafael wakes up in the morning, and his phone presents a notification that his printer
342 attempted to install new Firmware at the chosen time, but the update process failed. The
343 phone presents a notification with a link to an assistance page.

3.4 Out of Scope

The following are considered out of scope for this specification:

1. Definition of Firmware file formats;
2. Supporting updating Printer with Firmware from an arbitrary URI; and
3. Special states a Printer is required to be in to receive and/or install a Firmware update.

3.5 Design Requirements

The design requirements for this specification are:

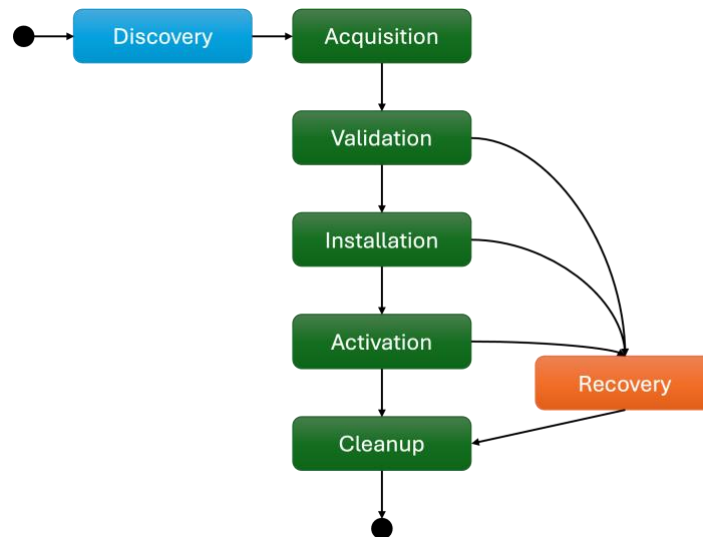
1. Define attributes and values to describe discovered new Firmware available from the configured Firmware Repository;
2. Define operations for requesting the Printer update its Firmware;
3. Define security and privacy requirements necessary to support these operations and attributes; and
4. Define sections to register all attributes, values, and operations with IANA.

4. Model

This specification extends the IPP model to support the Triggered Autonomous Firmware Update scenario described below.

The ability to update a device's Firmware has gone from being a useful but optional feature to an essential feature that is increasingly mandated by cybersecurity regulations. Figure 1 describes the phases involved in performing a Firmware Update.

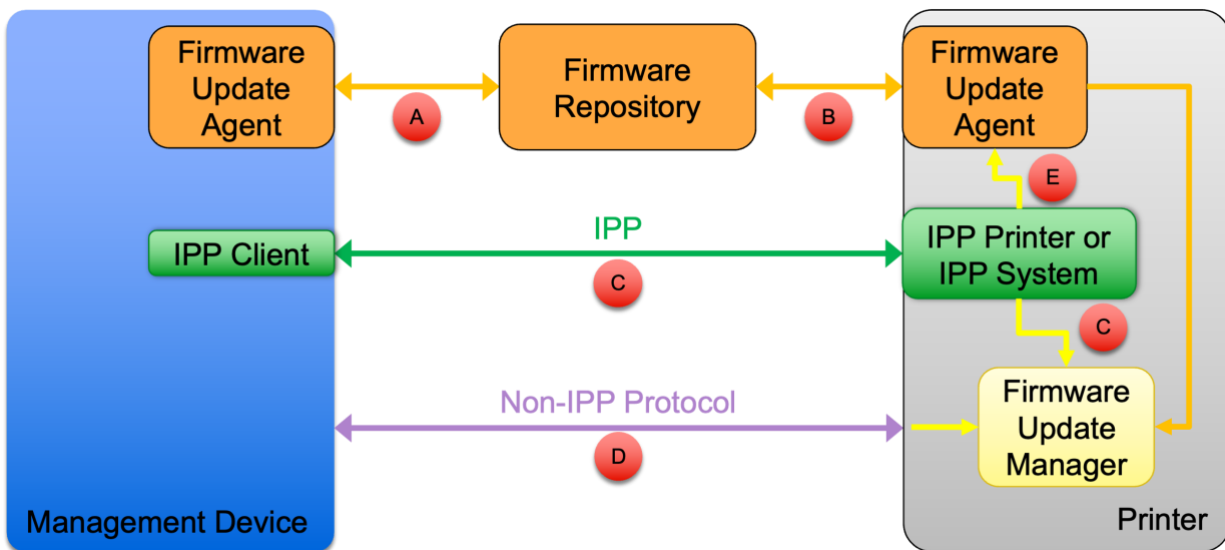
Figure 1 - Firmware Update Phases



4.1 Firmware Update Interaction Types

Figure 2 provides an illustration describing various entities that can participate in each of those phases. a high-level conceptual model for various means of installing new Firmware in a Printer.

Figure 2 - Firmware Update Conceptual Model



Several different interactions can be derived from this model. Each is described in the following subsections.

4.1.1 Directed Firmware Update

A Firmware Update Agent running on a Management Device performs the Discovery phase by checking a Firmware Repository (A) for available new Firmware. The Management Device executes the Acquisition phase by downloading the new Firmware from the Firmware Repository, possibly performs the Validation phase, and then completes the Acquisition phase by submitting it to the Printer. If the Printer implements the IPP System Service [PWG 5100.22] then the Management Device can submit the new Firmware via IPP (C). The Printer and Management Device might additionally implement some other protocol and system for completing the Acquisition phase (D). The Printer might perform its own Validation, and then performs the Installation, Activation and Cleanup phases.

4.1.2 Autonomous Firmware Update

A Firmware Update Agent running in the Printer itself executes all phases. It performs the Discovery phase by checking for available new Firmware from its Firmware Repository (B). If new Firmware is available, the Printer's Firmware Update Manager performs the Acquisition, Validation, Installation, Activation and Cleanup phases by downloading the new Firmware, validating that it is legitimate and unaltered, installs the new Firmware, activates it (possibly by restarting the host device), and then cleans up. The Printer can indicate its progress and status installing new Firmware via IPP (C). All these phases are executed within the Printer itself, without any user interaction.

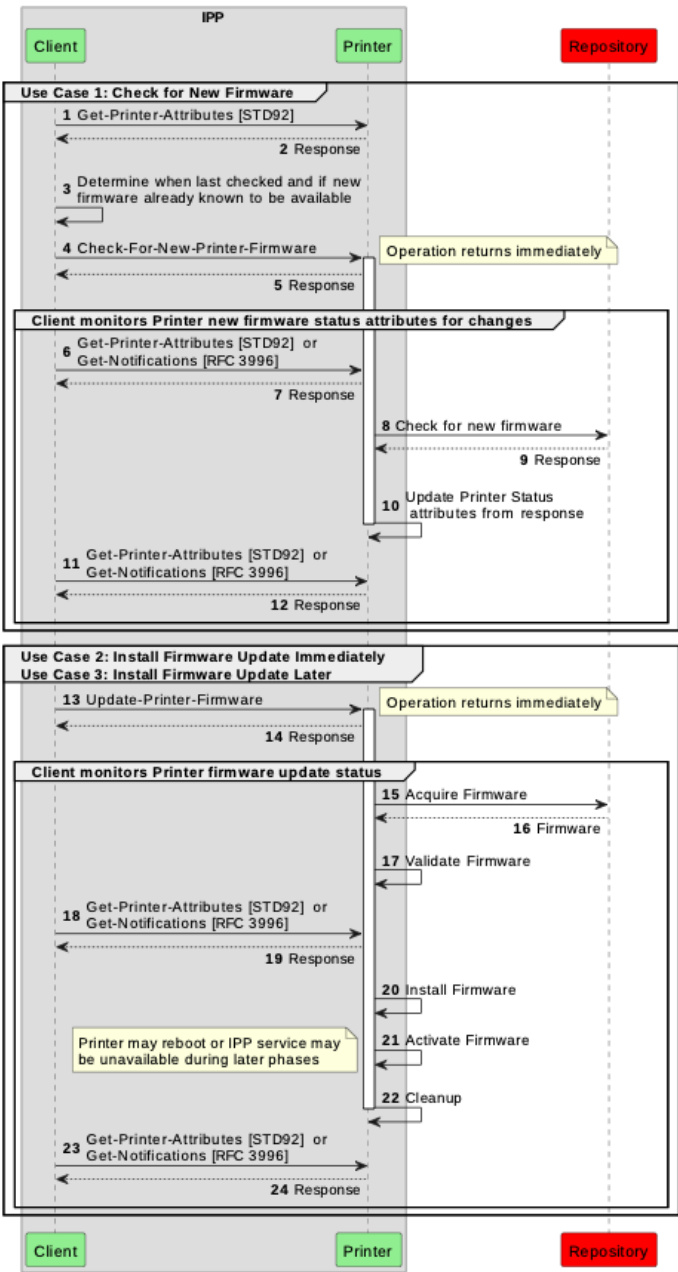
4.1.3 Triggered Autonomous Firmware Update

This model provides IPP operations and attributes to allow an IPP Client to request via IPP that the Printer perform the Discovery phase and the Acquisition / Validation / Installation / Activation / Cleanup set of phases described in the Autonomous Firmware Update. An IPP Client requests (C) the IPP System or IPP Printer to cause the Printer's Firmware Update Agent to execute the Discovery phase (E). If new Firmware is discovered to be available, the Printer reports information about available new Firmware via IPP. The IPP Client can also request the Printer to execute the Acquisition / Validation / Installation / Activation / Cleanup phases, either immediately or at a future time indicated in the request. The Printer can indicate its progress and status installing new Firmware via IPP (C).

This specification defines the new "Check-For-New-Printer-Firmware" IPP operation (section 5.1) to provide access to the Printer's Discovery phase, and also defines the "Update-Printer-Firmware" IPP operation (section 5.2) to provide IPP access to cause the execution of the Acquisition / Validation / Installation / Activation / Cleanup phases to support the "Triggered Autonomous Firmware Update" interaction model, as illustrated in Figure 3.

411

Figure 3 - Triggered Autonomous Firmware Update Sequence Diagram



412

413 **4.2 New Firmware Available Status**

414 A Printer reports when it last checked for new Firmware with its “printer-new-firmware-

415 check-date-time” attribute (section 6.3.1).

416 If a Printer has attempted to discover whether new Firmware is available from its Firmware

417 Update Repository, it indicates the result of the most recent attempt using one of the new

418 keywords defined in this specification for the “printer-state-reasons” attribute (section 7.1).

If new Firmware has been discovered, then Printer adds the 'new-firmware-available' keyword (section 7.3.4) to its "printer-state-reasons" attribute and describes the available new Firmware using the attributes listed in Table 1. This suite of attributes was defined to correspond to the "firmware-xxx" attributes [PWG 5100.13].

Table 1 - New Firmware Status Attributes

Attribute	Section	Current Firmware Status Attribute
"printer-new-firmware-check-date-time"	6.3.1	n/a
"printer-new-firmware-name"	6.3.5	"printer-firmware-name"
"printer-new-firmware-patches"	6.3.8	"printer-firmware-patches"
"printer-new-firmware-k-octets"	6.3.6	n/a
"printer-new-firmware-string-version"	6.3.9	"printer-firmware-string-version"
"printer-new-firmware-support-uri"	6.3.5	n/a
"printer-new-firmware-urgency"	6.3.10	n/a
"printer-new-firmware-version"	6.3.11	"printer-firmware-version"

If the Printer detects that no new Firmware is available, the Printer removes the 'new-firmware-available' keyword from its "printer-state-reasons" attribute and reports the defined out-of-band values for the attributes listed in Table 1.

4.3 Requesting Printer to Check for New Firmware

A Client sends the Check-For-New-Firmware-Update operation (section 5.1) to request the Printer to check for new Firmware. The Printer responds immediately before completing the check. A Client can use the Get-Printer-Attributes [STD92] or Get-Notifications [RFC 3996] operations and the attributes listed in section 4.2 to discover the availability of new Firmware.

To protect itself and its Firmware Update Repository from Denial of Service (DoS) attacks, the Printer can provide cached values for the attributes listed in Table 1 for a reasonable amount of time. Additional guidance is provided in section 10.

4.4 Requesting Printer to Update Firmware

A Client sends the Update-Printer-Firmware operation (section 5.2) to request the Printer to install new Firmware if available. The request can include the "delay-update-until" attribute (section 6.1.1) or the "delay-update-until-date-time" attribute (section 6.1.2) to schedule when the Printer installs the available new Firmware. The Printer may download available new Firmware at any time.

A Client can monitor the Printer's progress and status using the Get-Printer-Attributes [STD92] or Get-Notifications [RFC 3996] operations and examining the Printer's "printer-state-reasons" attribute for keywords listed in section 7.3.

445 Note: The Printer can reboot during the process of installing new Firmware, causing the
446 Printer to be temporarily unreachable.

447 **4.5 Selecting a Printer Firmware Repository**

448 The “printer-firmware-repository-uri-configured” attribute (section 6.2.4) reports the URI of
449 the Printer’s currently configured Firmware Repository. The “printer-firmware-repository-
450 uri-supported” attribute (section 6.2.5) lists the set of Firmware Repository URIs from
451 which the “printer-firmware-repository-uri-configured” can be set.

452 The Printer’s “printer-settable-attributes-supported” attribute [RFC 3380] will list ‘printer-
453 firmware-repository-uri-configured’ and/or ‘printer-firmware-repository-uri-supported’ if
454 either or both of those attributes can be configured using IPP via a Set-Printer-Attributes
455 operation [RFC 3380]. An authorized Client can select a different Firmware Repository
456 using the Set-Printer-Attributes operation to set the value of “printer-firmware-repository-
457 uri-configured” attribute to one of the values listed by “printer-firmware-repository-uri-
458 supported”. An authorized Client can configure which Firmware Repositories are allowed
459 by setting the “printer-new-firmware-update-repository-supported” attribute using the Set-
460 Printer-Attributes operation.

461 An Authorized Client can use the Get-Printer-Supported-Values operation [RFC 3380] to
462 obtain the factory default list of values for the “printer-firmware-repository-uri-supported”
463 attribute.

464 **4.6 Updating the Printer Firmware Update Feature Policy**

465 The “printer-new-firmware-policy-configured” attribute (section 6.2.2) attribute reports the
466 Printer’s currently configured policy for checking for new Firmware. The “printer-new-
467 firmware-policy-supported” attribute (section 6.2.3) lists the set of policies supported by the
468 Printer.

469 The Printer’s “printer-settable-attributes-supported” attribute [RFC 3380] will list ‘printer-
470 new-firmware-policy-configured’ and/or ‘printer-new-firmware-policy-supported’ if either or
471 both of those attributes can be configured using IPP via a Set-Printer-Attributes operation
472 [RFC 3380]. An authorized Client can configure the Printer’s policy for whether it
473 automatically checks for, downloads, and/or installs new Firmware by setting the value of
474 the “printer-new-firmware-policy-configured” attribute using the Set-Printer-Attributes
475 operation [RFC 3380].

5. New Operations

5.1 Check-For-New-Printer-Firmware

This RECOMMENDED operation causes the Printer to check its Firmware repository for available new Firmware. The Printer responds immediately before performing the check. If new Firmware is available, it indicates the results as described in section 4.2.

The Authenticated User performing this operation MUST be either an Operator or Administrator for the Printer. Otherwise, the Printer MUST reject the operation and return the 'client-error-forbidden', 'client-error-not-authenticated', or 'client-error-not-authorized' status code, as appropriate.

5.1.1 Check-For-New-Printer-Firmware Request

The following groups of attributes are supplied in a Check-For-New-Printer-Firmware request:

Group 1: Operation Attributes

"attributes-charset" (charset) and "attributes-natural-language" (naturalLanguage):

The Client MUST supply and the Printer MUST support both of these attributes, as described in [STD92] Section 4.1.4.1.

"printer-uri" (uri):

The Client MUST supply and the Printer MUST support this attribute, which specifies the target for this operation, as described in [STD92] Section 4.1.5.

"requesting-user-name" (name(MAX)):

The Client SHOULD supply and the Printer MUST support this attribute, as described in [STD92] Section 9.3.

"requesting-user-uri" (uri):

The Client SHOULD supply and the Printer MUST support this attribute, as described in [PWG 5100.13] section 5.1.6.

5.1.2 Check-For-New-Printer-Firmware Response

The following groups of attributes are supplied in a Check-For-New-Printer-Firmware response:

Group 1: Operation Attributes

505 "attributes-charset" (charset) and "attributes-natural-language" (naturalLanguage):

506 The Client MUST support and the Printer MUST supply both of these attributes, as
507 described in [STD92] Section 4.1.4.1.

508 Status Message:

509 In addition to the REQUIRED status-code returned, the response MAY include the
510 "status-message" (text(255)) and/or the "detailed-status-message" (text(MAX))
511 operation attribute as described in [STD92] Appendix B and Section 4.1.6.

512 Group 2: Printer Attributes

513 "printer-new-firmware-last-checked" (dateTime)

514 The Printer MAY supply this attribute, as defined in section 6.3.1.

515 "printer-new-firmware-k-octets" (integer(1:MAX) | unknown)

516 The Printer MAY supply this attribute, as defined in section 6.3.6.

517 "printer-new-firmware-support-uri" (uri)

518 The Printer MAY supply this attribute, as defined in section 6.3.5.

519 "printer-new-firmware-urgency"

520 The Printer MAY supply this attribute, as defined in section 6.3.10

521 "printer-new-firmware-version"

522 The Printer MAY supply this attribute, as defined in section 6.3.11

523 "printer-new-firmware-string-version"

524 The Printer MAY supply this attribute, as defined in section 6.3.9

525 **5.2 Update-Printer-Firmware**

526 This REQUIRED operation causes the Printer to attempt to perform a Firmware self-
527 update if Firmware is available from the currently configured Firmware Repository that is
528 newer than what is currently installed. The Printer responds immediately, without checking
529 the reachability of the configured Firmware Repository or attempting to begin the Firmware
530 update.

531 If the operation is successful and the Client supplies either the "delay-update-until" or
532 "delay-update-until-time" operation attributes, then the Printer does not begin performing
533 the Firmware update until the specified time, and includes the current "firmware-update-

delayed-until" (section 6.3.1) and "firmware-update-delayed-until-time" (section 6.3.4) attributes in the response. If new Firmware installation was already scheduled with a delay, and changes to that are desired, then a Client can perform a new Update-Printer-Firmware request to supply a new value.

A Client MUST NOT supply both the "delay-update-until" and "delay-update-until-time" (section 6.1.2) operation attributes in an Update-Printer-Firmware request. If a Client supplies both attributes, then the Printer MUST reject the request and return the 'client-error-conflicting-attributes' status code. While the specified time period has not yet arrived, the Printer includes the 'firmware-update-delay-until-specified' value in the Printer's "printer-state-reasons" Printer Status attribute (section 7.1).

When the time period arrives, the Printer sets its "printer-state" attribute to 5 ('stopped'), sets the value of its "printer-is-accepting-jobs" Printer Status attribute [STD92] to 'false', adds the "firmware-update-in-progress" keyword to its "printer-state-reasons" Printer Status attribute (section 7.1), and begins the Firmware update. When the Firmware update activity concludes, the Printer indicates the result by removing the 'firmware-update-in-progress' keyword from its "printer-state-reasons" Printer Status attribute, setting the value of its "printer-is-accepting-jobs" attribute to 'true', and setting its "printer-state" attribute to 3 ('idle'). The Printer also adds either the 'firmware-update-success' or 'firmware-update-failure' keyword, as appropriate, to its "printer-state-reasons" attribute, for a duration that is implementation specific. If the Firmware update is successful, then the Printer clears the values for its "printer-new-firmware-last-checked" (section 6.3.1), "printer-new-firmware-name" (section 6.3.5), "printer-new-firmware-patches" (section 6.3.8), "printer-new-firmware-k-octets" (section 6.3.6), "printer-new-firmware-string-version" (section 6.3.9), "printer-new-firmware-support-uri" (section 6.3.5), "printer-new-firmware-urgency" (section 6.3.10), and "printer-new-firmware-version" (section 6.3.11) attributes.

The Client sends the Update-Printer-Firmware request over a secure connection [STD92]. If the Printer receives an Update-Printer-Firmware request over an insecure connection (e.g. HTTP without TLS), then the Printer MUST respond with an HTTP 426 response to upgrade the connection to TLS [RFC 2817] [RFC9110] or reject the request.

The Authenticated User performing this operation MUST be an Operator or Administrator for the Printer. Otherwise, the Printer MUST reject the operation and return the 'client-error-forbidden', 'client-error-not-authenticated', or 'client-error-not-authorized' status code, as appropriate.

5.2.1 Update-Printer-Firmware Request

The following groups of attributes are supplied in an Update-Printer-Firmware request:

Group 1: Operation Attributes

"attributes-charset" (charset) and "attributes-natural-language" (naturalLanguage):

571 The Client MUST supply and the Printer MUST support both of these attributes, as
572 described in [STD92] Section 4.1.4.1.

573 "printer-uri" (uri):

574 The Client MUST supply and the Printer MUST support this attribute, which
575 specifies the target for this operation, as described in [STD92] Section 4.1.5.

576 "requesting-user-name" (name(MAX)):

577 The Client MAY supply and the Printer MUST support this attribute, as described in
578 [STD92] Section 9.3.

579 "delay-update-until" (type2 keyword | name(MAX))

580 The Client MAY supply and the Printer MAY support this attribute, as defined in
581 section 6.1.1.

582 "delay-update-until-time" (dateTime)

583 The Client MAY supply and the Printer MAY support this attribute, as defined in
584 section 6.1.2.

585 **5.2.2 Update-Printer-Firmware Response**

586 The following groups of attributes are supplied in an Update-Printer-Firmware response:

587 Group 1: Operation Attributes

588 "attributes-charset" (charset) and "attributes-natural-language" (naturalLanguage):

589 The Client MUST support and the Printer MUST supply both of these attributes, as
590 described in [STD92] Section 4.1.4.1.

591 Status Message:

592 In addition to the REQUIRED status-code returned, the response MAY include the
593 "status-message" (text(255)) and/or the "detailed-status-message" (text(MAX))
594 operation attribute as described in [STD92] Appendix B and Section 4.1.6.

595 Group 2: Printer Attributes

596 "printer-new-firmware-delayed-until" (type2 keyword)

597 The Printer MAY supply this attribute, as described in section 6.3.1.

598 "printer-new-firmware-delayed-until-time" (dateTime | no-value)

599 The Printer MAY supply this attribute, as described in section 6.3.4.

Group 3: Unsupported Attributes

See [STD92] Section 4.1.7 for details on returning unsupported attributes.

6. New Attributes

6.1 Operation Attributes

6.1.1 delay-update-until (type2 keyword)

This attribute specifies a named time period when the Printer will install new Firmware as requested in an Update-Printer-Firmware operation (section 5.2). Table 2 lists the keyword values defined in this specification for use with this attribute.

Table 2 – Keywords for the "delay-update-until" attribute

Keyword	Description
'no-delay'	Install new Firmware immediately.
'indefinite'	Delay new Firmware install indefinitely.
'day-time'	Delay new Firmware install until daylight.
'evening'	Delay new Firmware install until the evening.
'night'	Delay new Firmware install until the night.
'second-shift'	Delay new Firmware install until the second work shift.
'third-shift'	Delay new Firmware install until the third work shift.
'weekend'	Delay new Firmware install until the weekend.

6.1.2 delay-update-until-date-time (dateTime)

This attribute specifies a precise date and time when the Printer will install the new Firmware as requested in an Update-Printer-Firmware operation (section 5.2).

6.2 Printer Description Attributes

6.2.1 delay-update-until-supported (1setOf type2 keyword)

This REQUIRED attribute indicates the values the Printer supports for the “delay-update-until” attribute (section 6.1.1).

6.2.2 printer-new-firmware-policy-configured (type2 keyword)

This REQUIRED attribute indicates the currently configured policy for the Triggered Autonomous Firmware Update feature (section 4). Table 3 lists the keywords defined in this specification.

If the Printer supports configuring this attribute via IPP, then it will list this attribute in its “printer-settable-attributes-supported” attribute [RFC 3380].

Table 3 – Keywords for "printer-new-firmware-policy-supported"

Keyword	Description
'auto'	Printer automatically discovers, downloads, validates, installs and activates new Firmware
'check'	Printer discovers new Firmware but does not download / validate / install / activate it
'directed'	Printer discovers and validates / installs / activates new Firmware using the operations defined in section 5
'download'	Printer automatically discovers, downloads and validates new Firmware but refrains from installing it

6.2.3 printer-new-firmware-policy-supported (1setOf type2 keyword)

This REQUIRED attribute lists the supported policies for Triggered Autonomous Firmware Update (section 4). If the Printer supports configuring this attribute via IPP, then it will list this attribute in its "printer-settable-attributes-supported" attribute [RFC 3380].

6.2.4 printer-firmware-repository-uri-configured (uri)

This CONDITIONALLY REQUIRED attribute provides the currently configured URI of the Firmware Repository the Printer uses to check for new Firmware, either on its own or in response to a Check-For-New-Printer-Firmware operation (section 5.1).

If the Printer supports changing this attribute's value, then the Printer MUST list this attribute's name in its "printer-settable-attributes-supported" attribute [RFC 3380], MUST support the Set-Printer-Attributes operation [RFC 3380], MUST support the "printer-firmware-repository-uri-supported" attribute (section 6.2.5), and MUST only accept a value that listed by the Printer's "printer-firmware-repository-uri-supported" attribute.

6.2.5 printer-firmware-repository-uri-supported (1setOf uri)

This attribute lists the Firmware Repository URIs that can be used to configure the Printer's "printer-firmware-repository-uri-configured" attribute (section 6.2.2).

If the Printer supports changing this attribute's values, then the Printer MUST support the Get-Printer-Supported-Values operation [RFC 3380] and the value returned for this attribute MUST include the manufacturer default Firmware Repository URI.

6.3 Printer Status Attributes

6.3.1 printer-firmware-info-uri (uri)

This REQUIRED attribute provides a 'http:' or 'https:' scheme URI to user readable content describing the currently installed Firmware.

6.3.2 printer-new-firmware-check-date-time (dateTime | no-value | unknown)

This REQUIRED attribute reports the value reported by the “printer-current-time” attribute [STD92] when the Printer last attempted to check for new Firmware, either on its own or in response to a Check-For-New-Printer-Firmware operation (section 5.1). The value is updated even when the Printer is unable to contact the configured Firmware Repository or obtain actionable information. If the Printer has never checked for new Firmware, then the Printer MUST return the ‘no-value’ out-of-band value.

Note: If the Printer’s clock is not set when the new Firmware check is run, then the Printer’s “printer-current-time” attribute [STD92] will report the ‘unknown’ out-of-band value, and in such circumstances this attribute will also report ‘unknown’.

6.3.3 printer-new-firmware-delayed-until (type2 keyword)

This attribute indicates the delay set from the "delay-update-until" operation attribute (section 6.1.1) received in a previous Update-Printer-Firmware operation request (section 5.2) that was successful and accepted by the Printer. Table 2 lists the keyword values defined in this specification this attribute can hold. A Printer will omit this attribute if the most recent Update-Printer-Firmware request was successful and did not include “delay-update-until”.

6.3.4 printer-new-firmware-delayed-until-date-time (dateTime)

This attribute indicates the delay set from the "delay-update-until-date-time" operation attribute (section 6.1.2) received in a previous Update-Printer-Firmware operation request (section 5.2) that was successful and accepted by the Printer. A Printer will omit this attribute if the most recent Update-Printer-Firmware request was successful and did not include “delay-update-until-date-time”.

6.3.5 printer-new-firmware-info-uri (uri | no-value | unknown)

This REQUIRED attribute provides a 'http:' or 'https:' URI to user readable content describing the available new Firmware. If the Printer has never checked for new Firmware, then the Printer returns the ‘no-value’ out-of-band value. If the Printer was unable to contact the configured Firmware Repository or unable to get actionable information from it, then the Printer returns the ‘unknown’ out-of-band value.

6.3.6 printer-new-firmware-k-octets (1setOf integer(1:MAX) | no-value | unknown)

This REQUIRED attribute indicates the size of the new Firmware, measured in kiB (1024 octets). If the size is unknown, then the Printer returns the 'unknown' out-of-band value. If the Printer has never checked for new Firmware, then the Printer returns the ‘no-value’ out-of-band value. If the Printer was unable to contact the configured Firmware Repository or unable to get actionable information from it, then the Printer returns the ‘unknown’ out-of-band value.

6.3.7 printer-new-firmware-name (1setOf name(MAX) | no-value | unknown)

This REQUIRED attribute provides the names of the new Firmware, in a format matching the Printer's "printer-firmware-name" attribute [PWG 5100.13]. If the Printer has never checked for new Firmware, then the Printer returns the 'no-value' out-of-band value. If the Printer was unable to contact the configured Firmware Repository or unable to get actionable information from it, then the Printer returns the 'unknown' out-of-band value.

6.3.8 printer-new-firmware-patches (1setOf text(MAX) | no-value | unknown)

This REQUIRED attribute provides the set of patches to be applied to each of the Printer firmware components named by the "printer-new-firmware-name" Printer Status attribute, in a format matching the Printer's "printer-firmware-patches" attribute [PWG 5100.13]. If the Printer has never checked for new Firmware, then the Printer returns the 'no-value' out-of-band value. If the Printer was unable to contact the configured Firmware Repository or unable to get actionable information from it, then the Printer returns the 'unknown' out-of-band value.

6.3.9 printer-new-firmware-string-version (1setOf text(MAX) | no-value | unknown)

This REQUIRED attribute provides the string versions of the values provided by the "printer-new-firmware-version" attribute (section 6.3.11), in a format matching the Printer's "printer-firmware-string-version" attribute [PWG 5100.13]. If the Printer has never checked for new Firmware, then the Printer returns the 'no-value' out-of-band value. If the Printer was unable to contact the configured Firmware Repository or unable to get actionable information from it, then the Printer returns the 'unknown' out-of-band value.

6.3.10 printer-new-firmware-urgency (type2 keyword | no-value | unknown)

This REQUIRED attribute indicates the urgency of the new Firmware. Table 4 lists the keywords defined for this attribute. If the Printer has never checked for new Firmware, then the Printer returns the 'no-value' out-of-band value. If the Printer was unable to contact the configured Firmware Repository or unable to get actionable information from it, then the Printer returns the 'unknown' out-of-band value.

Table 4 – Keywords for "printer-new-firmware-urgency"

Keyword	Description
'bugfix'	The available new Firmware provides minor bug fixes
'feature'	The available new Firmware provides new features
'security'	The available new Firmware provides security updates

6.3.11 printer-new-firmware-version (1setOf octetString(MAX) | no-value | unknown)

This REQUIRED attribute reports the latest firmware version available for the Printer by the Printer's manufacturer, in a format matching the Printer's "printer-firmware-version" attribute [PWG 5100.13]. If the Printer has never checked for new Firmware, then the

Printer returns the 'no-value' out-of-band value. If the Printer was unable to contact the configured Firmware Repository or unable to get actionable information from it, then the Printer returns the 'unknown' out-of-band value.

7. New Values for Existing Attributes

7.1 ipp-features-supported

This specification defines the new 'fwupdate' keyword for the "ipp-features-supported" attribute [PWG 5100.13] to indicate support for this specification. If the Printer supports the "ipp-features-supported" attribute, then the Printer SHOULD include the "fwupdate" keyword among the keywords listed by that attribute.

7.2 operations-supported (1setOf type2 enum)

Table 5 lists the following new keywords for the "operations-supported" attribute [STD92] defined by this specification.

Table 5 – Enum values for "operations-supported"

Enum	Description
0x006B	Check-For-New-Printer-Firmware
0x006C	Update-Printer-Firmware

7.3 printer-state-reasons (1setOf type2 keyword)

This section defines new keywords for the "printer-state-reasons" attribute [STD92]. These are used to indicate what phase of the Firmware installation process is occurring currently, as described in Figure 1. Keywords are defined for each phase to indicate if the phase is "in progress", ended in "failure" or ended in "success". Other keywords indicate specific failure conditions that benefit from more detail or other status conditions possibly occurring at other times.

7.3.1 firmware-repository-access-error

This REQUIRED keyword indicates that the Firmware Repository is reachable but that the new Firmware could not be downloaded. Possible reasons could include TLS server certificate rejection, cipher suite rejection, missing file, or unauthorized / forbidden access.

7.3.2 firmware-repository-unreachable

This REQUIRED keyword indicates that the Printer was unable to connect to the currently configured Firmware Repository as specified by the Printer's "printer-firmware-repository-uri-configured" attribute (section 6.2.4).

742 7.3.3 new-firmware-installation-delay-until-specified

743 This REQUIRED keyword indicates that the Printer received an Update-Printer-Firmware
744 request (section 5.2.1) that included either the “delay-update-until” (section 6.1.1)
745 operation attribute or “delay-update-until-date-time” operation attribute (section 6.1.2).

746 When this keyword is listed by the “printer-state-reasons” attribute, details of the delay will
747 be indicated by the Printer’s “printer-new-firmware-delayed-until” attribute (section 6.3.3) or
748 “printer-new-firmware-delayed-until-date-time” attribute (section 6.3.4).

749 7.3.4 new-firmware-available

750 This REQUIRED keyword indicates there is new Firmware available.

751 7.3.5 new-firmware-acquisition-failure

752 This REQUIRED keyword indicates that the Printer’s most recent attempt to perform the
753 Acquisition phase of installing new Firmware has failed.

754 7.3.6 new-firmware-acquisition-in-progress

755 This REQUIRED keyword indicates that the Printer is currently performing the Acquisition
756 phase of installing new Firmware.

757 7.3.7 new-firmware-acquisition-success

758 This REQUIRED keyword indicates that the Printer’s most recent attempt to perform the
759 Acquisition phase of installing new Firmware has succeeded.

760 7.3.8 new-firmware-activation-failure

761 This REQUIRED keyword indicates that the Printer’s most recent attempt to perform the
762 Activation phase of installing new Firmware has failed.

763 7.3.9 new-firmware-activation-in-progress

764 This REQUIRED keyword indicates that the Printer is currently performing the Activation
765 phase of installing new Firmware.

766 7.3.10 new-firmware-activation-success

767 This REQUIRED keyword indicates that the Printer’s most recent attempt to perform the
768 Activation phase of installing new Firmware has succeeded.

769 7.3.11 new-firmware-cleanup-failure

770 This REQUIRED keyword indicates that the Printer’s most recent attempt to perform the
771 Cleanup phase of installing new Firmware has failed.

772 7.3.12 new-firmware-cleanup-in-progress

773 This REQUIRED keyword indicates that the Printer is currently performing the Cleanup
774 phase of installing new Firmware.

775 7.3.13 new-firmware-cleanup-success

776 This REQUIRED keyword indicates that the Printer's most recent attempt to perform the
777 Cleanup phase of installing new Firmware has succeeded.

778 7.3.14 new-firmware-installation-failure

779 This REQUIRED keyword indicates that the Printer's most recent attempt to perform the
780 Installation phase of installing new Firmware has failed.

781 7.3.15 new-firmware-installation-in-progress

782 This REQUIRED keyword indicates that the Printer is currently performing the Install
783 phase of installing new Firmware.

784 7.3.16 new-firmware-installation-success

785 This REQUIRED keyword indicates that the Printer's most recent attempt to perform the
786 Install phase of installing new Firmware has succeeded.

787 7.3.17 new-firmware-recovery-failure

788 This REQUIRED keyword indicates that the Printer's most recent attempt to perform the
789 Recovery phase of installing new Firmware has failed.

790 7.3.18 new-firmware-recovery-in-progress

791 This REQUIRED keyword indicates that the Printer is currently performing the Recovery
792 phase of installing new Firmware by reactivating the previously installed Firmware
793 following a problem with either the Installation or Activation phases.

794 7.3.19 new-firmware-recovery-success

795 This REQUIRED keyword indicates that the Printer's most recent attempt to perform the
796 Recovery phase of installing new Firmware has succeeded.

797 7.3.20 new-firmware-validation-failure

798 This REQUIRED keyword indicates that the Printer's most recent attempt to perform the
799 Validation phase of installing new Firmware has failed.

7.3.21 new-firmware-validation-in-progress

This REQUIRED keyword indicates that the Printer is currently performing the Validation phase of installing new Firmware.

7.3.22 new-firmware-validation-success

This REQUIRED keyword indicates that the Printer's most recent attempt to perform the Validation phase of installing new Firmware has succeeded.

8. Conformance Requirements

8.1 Printer Conformance Requirements

Printers conforming to this Specification MUST support:

1. The model described in section 4.1.3;
2. The required operations defined in section 5;
3. The required attributes and values defined in section 6;
4. The required additional values defined in section 7;
5. The internationalization considerations defined in section 9; and
6. The security and privacy considerations defined in section 10.

8.2 Client Conformance Requirements

Clients conforming to this Specification MUST support:

1. The model described in section 4.1.3;
2. The required operations defined in section 5;
3. The required attributes and values defined in section 6;
4. The required additional values defined in section 7;
5. The internationalization considerations defined in section 9; and
6. The security and privacy considerations defined in section 10.

9. Internationalization Considerations

For interoperability and basic support for multiple languages, conforming implementations MUST support:

1. The Universal Character Set (UCS) Transformation Format -- 8 bit (UTF-8) [STD63] encoding of Unicode [UNICODE] [ISO10646]; and
2. The Unicode Format for Network Interchange [RFC 5198] which requires transmission of well-formed UTF-8 strings and recommends transmission of normalized UTF-8 strings in Normalization Form C (NFC) [UAX15].

Unicode NFC is defined as the result of performing Canonical Decomposition (into base characters and combining marks) followed by Canonical Composition (into canonical composed characters wherever Unicode has assigned them).

WARNING – Performing normalization on UTF-8 strings received from Clients and subsequently storing the results (e.g., in Job objects) could cause false negatives in Client searches and failed access (e.g., to Printers with percent-encoded UTF-8 URIs now 'hidden').

Implementations of this specification SHOULD conform to the following standards on processing of human-readable Unicode text strings, see:

- Unicode Bidirectional Algorithm [UAX9] – left-to-right, right-to-left, and vertical
- Unicode Line Breaking Algorithm [UAX14] – character classes and wrapping
- Unicode Normalization Forms [UAX15] – especially NFC for [RFC 5198]
- Unicode Text Segmentation [UAX29] – grapheme clusters, words, sentences
- Unicode Identifier and Pattern Syntax [UAX31] – identifier use and normalization
- Unicode Collation Algorithm [UTS10] – sorting
- Unicode Locale Data Markup Language [UTS35] – locale databases

Implementations of this specification are advised to also review the following informational documents on processing of human-readable Unicode text strings:

- Unicode Character Encoding Model [UTR17] – multi-layer character model
- Unicode Character Property Model [UTR23] – character properties
- Unicode Conformance Model [UTR33] – Unicode conformance basis

10. Security and Privacy Considerations

Provide security and privacy considerations for this specification, such as the following.

The IPP extensions defined in this document require the same security and privacy considerations as defined in the Internet Printing Protocol/1.1 [STD92].

Implementations of this specification SHOULD conform to the following standard on processing of human-readable Unicode text strings, see:

- Unicode Security Mechanisms [UTS39] – detecting and avoiding security attacks

859 • Implementations of this specification are advised to also review the following
860 informational document on processing of human-readable Unicode text strings:

861 • Unicode Security FAQ [UNISECFAQ] – common Unicode security issues

862 Implementations of this specification SHOULD comply with the following:

863 • NIST SP800-193 Platform Firmware Resiliency Guidelines (May 2018)
864 <https://csrc.nist.gov/pubs/sp/800/193/final>

865 • ITU-T X.1373 Secure Software Update Capability for ITS (March 2024)
866 <https://www.itu.int/ITU-T/recommendations/rec.aspx?rec=15664>

867 Printers implementing this specification SHOULD comply with the following guidance:

868 1. Communications between the Printer and the Firmware Repository is protected in
869 transit (i.e., TLS, DTLS, etc.);

870 2. Firmware files are cryptographically signed, and Printers validate these
871 cryptographic signatures before proceeding with installation;

872 3. Identifiers used to track information about existing Firmware versions, authorize
873 access to new Firmware, and/or monitor download and installation performance do
874 not include Personally Identifiable Information (PII).

875 **11. Implementation and Deployment Considerations**

876 This specification defines only an IPP interface to allow a Client to interact with a subset of
877 a Printer's own Firmware update capability. Though this helps to facilitate a Firmware
878 update user experience using IPP, other elements of the user experience are the sole
879 responsibilities of the Client, the Printer and the Firmware Repository.

880 A well-implemented Client supporting this specification SHOULD:

881 1. Present UI elements appropriately adjacent to other Printer basic information that
882 lists the Printer's "printer-firmware-xxx" attributes defined in [PWG 5100.13] and this
883 specification, including the "printer-firmware-info-uri" (section 6.3.1) so the user has
884 ready access to the release notes and other information about the currently installed
885 Firmware;

886 2. Update any cached Printer information with the latest settings, if caching is used, so
887 that the Printer's current capabilities and features are accurately reflected in the
888 Client's UI, and notification that the Firmware was updated;

889 3. Present UI elements that can appropriately present the Printer's status concerning
890 the current Firmware as well as any unsuccessful attempts to install available new
891 Firmware, and also provide UI elements to easily access the vendor's support page

for that Printer model, provided by the Printer’s “printer-support-uri” attribute (section **Error! Reference source not found.**).

A well-implemented Printer supporting this specification SHOULD:

1. Provide UI elements (physical or graphical) to indicate failed or incomplete installation of new Firmware on the Printer’s console;
2. Support the “printer-more-info-manufacturer” Printer Description attribute, with the value being a URI that points to the support page for the Printer’s particular model; and
3. Host a locally cached copy of the release notes for the currently installed Firmware.

A well-implemented Firmware Repository supporting this specification SHOULD provide the following:

1. Provide a “Release Notes” document describing the currently available new Firmware at a publicly accessible URL.

12. IANA Considerations

12.1 Attribute Registrations

The attributes defined in this specification will be published by IANA according to the procedures in the Internet Printing Protocol/1.1 [STD92] in the following file:

<https://www.iana.org/assignments/ipp-registrations>

The registry entries will contain the following information:

Operation Attributes:	Reference
-----	-----
delay-update-until (type2 keyword)	[FWUPDATE]
delay-update-until-date-time (dateTime)	[FWUPDATE]
Printer Description Attributes:	Reference
-----	-----
printer-new-firmware-policy-configured (type2 keyword)	[FWUPDATE]
printer-new-firmware-policy-supported (1setOf type2 keyword)	[FWUPDATE]
printer-firmware-repository-uri-configured (uri)	[FWUPDATE]
printer-firmware-repository-uri-supported (1setOf uri)	[FWUPDATE]
Printer Status Attributes:	Reference
-----	-----
printer-firmware-info-uri (uri)	[FWUPDATE]
printer-new-firmware-check-date-time (dateTime no-value unknown)	[FWUPDATE]


```

930 printer-new-firmware-delayed-until (type2 keyword | no-value)
931                                     [FWUPDATE]
932 printer-new-firmware-delayed-until-date-time (dateTime | no-value)
933                                     [FWUPDATE]
934 printer-new-firmware-info-uri (uri | no-value | unknown) [FWUPDATE]
935 printer-new-firmware-k-octets (1setOf integer(1:MAX) | no-value | unknown)
936                                     [FWUPDATE]
937 printer-new-firmware-name (1setOf name(MAX) | no-value | unknown)
938                                     [FWUPDATE]
939 printer-new-firmware-patches (1setOf text(MAX) | no-value | unknown)
940                                     [FWUPDATE]
941 printer-new-firmware-string-version (1setOf text(MAX) | no-value |
942 unknown) [FWUPDATE]
943 printer-new-firmware-urgency (type2 keyword | no-value | unknown)
944                                     [FWUPDATE]
945 printer-new-firmware-version (1setOf octetString(MAX) | no-value |
946 unknown) [FWUPDATE]
947

```

948 12.2 Type2 keyword Registrations

949 The attributes defined in this specification will be published by IANA according to the
 950 procedures in the Internet Printing Protocol/1.1 [STD92] in the following file:

951 <https://www.iana.org/assignments/ipp-registrations>

952 The registry entries will contain the following information:

Attributes (attribute syntax)	Reference
Keyword Attribute Value	
-----	-----
delay-update-until (type2 keyword)	[FWUPDATE]
no-hold	[FWUPDATE]
indefinite	[FWUPDATE]
day-time	[FWUPDATE]
evening	[FWUPDATE]
night	[FWUPDATE]
second-shift	[FWUPDATE]
third-shift	[FWUPDATE]
weekend	[FWUPDATE]
ipp-features-supported (1setOf type2 keyword)	[PWG5100.13]
fwupdate	[FWUPDATE]
printer-new-firmware-policy-configured (type2 keyword)	[FWUPDATE]
auto	[FWUPDATE]
check	[FWUPDATE]
directed	[FWUPDATE]
download	[FWUPDATE]
printer-new-firmware-delayed-until (type2 keyword no-value)	[FWUPDATE]
<Any "delay-update-until" keyword value>	[FWUPDATE]
printer-new-firmware-urgency (type2 keyword no-value unknown)	[FWUPDATE]
bugfix	[FWUPDATE]
feature	[FWUPDATE]

979	security	[FWUPDATE]
980	printer-state-reasons (type2 keyword)	[RFC8011]
981	firmware-repository-access-error	[FWUPDATE]
982	firmware-repository-unreachable	[FWUPDATE]
983	new-firmware-installation-delay-until-specified	[FWUPDATE]
984	new-firmware-available	[FWUPDATE]
985	new-firmware-acquisition-failure	[FWUPDATE]
986	new-firmware-acquisition-in-progress	[FWUPDATE]
987	new-firmware-acquisition-success	[FWUPDATE]
988	new-firmware-activation-failure	[FWUPDATE]
989	new-firmware-activation-in-progress	[FWUPDATE]
990	new-firmware-activation-success	[FWUPDATE]
991	new-firmware-installation-failure	[FWUPDATE]
992	new-firmware-installation-in-progress	[FWUPDATE]
993	new-firmware-installation-success	[FWUPDATE]
994		

995 12.3 Type2 enum Registrations

996 The attributes defined in this specification will be published by IANA according to the
 997 procedures in the Internet Printing Protocol/1.1 [STD92] in the following file:

998 <https://www.iana.org/assignments/ipp-registrations>

999 The registry entries will contain the following information:

1000	Attributes (attribute syntax)	
1001	Enum Value Enum Symbolic Name	Reference
1002	-----	-----
1003	operations-supported (1setOf type2 enum)	[RFC8011]
1004	0x006B Check-For-New-Printer-Firmware	[FWUPDATE]
1005	0x006C Update-Printer-Firmware	[FWUPDATE]

1006 12.4 Operation Registrations

1007 The attributes defined in this specification will be published by IANA according to the
 1008 procedures in the Internet Printing Protocol/1.1 [STD92] in the following file:

1009 <https://www.iana.org/assignments/ipp-registrations>

1010 The registry entries will contain the following information:

1011	Operation Name	Reference
1012	-----	-----
1013	Check-For-New-Printer-Firmware	[FWUPDATE]
1014	Update-Printer-Firmware	[FWUPDATE]

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The authors would also like to thank the following individuals for their contributions to this specification:

Bill Wagner
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15. Change History

This section will be removed when this document is published.

15.1 July 29, 2026

Updated with a few minor changes from the July 16 teleconference:

- Removed the “printer-support-uri” attribute from the 2026-07-16 draft, since “printer-more-info-manufacturer” [STD92] is already defined. Added “printer-more-info-manufacturer” to section 11;
- Updated IANA Considerations and section 7.3 to rename ‘xxx-download-xxx’ keywords to be ‘xxx-acquisition-xxx’;
- Added ‘xxx-activation-xxx’ keywords to section 7.3;
- Added a reference to RFC 9019; and
- Updated Section 11 to fix editorial issues.

15.2 July 16, 2026

Updated with changes from teleconference reviews and email feedback

- Changed the operation enum values for the 2 new operations because the ones listed previously were already listed in the draft for TRUSTNOONE;
- Added a new Firmware installation phase state diagram as Figure 1 in section 4;
- Added “-failure”, “-in-progress”, and “-success” keywords for each of the phases in the Figure 1;
- Fixed all the other points from the May F2F;
- Added “printer-firmware-info-uri” attribute, and renamed “printer-new-firmware-support-uri” to be “printer-new-firmware-info-uri”, to provide a URI to release notes, with more accurate attributes’ names;
- Added “printer-support-uri” to specifically point to the printer manufacturer’s model support page;
- Updated section 11 with implementation recommendations; and
- Update section 10 to add suggested additions from the mailing list.

1140 15.3 May 6, 2026

1141 Updated with some minor fixes identified in the April 23 teleconference, in advance of the
1142 May 2026 F2F:

- 1143 • Added a new “ipp-features-supported” keyword ‘fwupdate’ to indicate support for the
1144 Firmware Update feature;
- 1145 • Updated section 4 to include stages, and revised the text of section 4.3;
- 1146 • Changed several instances of “Firmware update” to instead be “available new
1147 Firmware” and similar editorial fixes; and
- 1148 • Fixed the IANA Considerations that had some wacky field cross-references that got
1149 out of control and otherwise updated to be current with the definitions earlier in the
1150 document, added in “printer-state-reasons”, and removed Status Codes section
1151 (thanks Mike!).

1152 15.4 April 22, 2026

1153 Updated with fixes to resolve issues identified in reviews from February 26 and March 26:

- 1154 • Revised names of a number of attributes and their signatures;
- 1155 • Updated section 4;
- 1156 • Made many small changes to make things consistent; large scale changes have
1157 calmed down; and
- 1158 • Updated language to normalize terminology for “new Firmware” vs. “Firmware
1159 upgrade”.

1160 15.5 February 25, 2026

1161 Updated as per feedback from February 2026 F2F:

- 1162 • Updated Abstract
- 1163 • Updated Introduction
- 1164 • Removed unused definitions in section 2
- 1165 • Updated section 4 including Figure 1 and added Table 1, and aligned on how to
1166 handle the attributes in Table 1

- 1167 • Added new “printer-new-firmware-policy-configured” and “printer-new-firmware-
1168 policy-supported” attributes, to support section 4.5

1169 **15.6 January 27, 2026**

1170 Several updates that provide a better structure now that we have consensus on the
1171 semantics of the operations

- 1172 • Updates to section 4 to consolidate discussion of the linkage between the new
1173 operations, the new attributes, and the new values for existing attributes
- 1174 • Updated the abstract
- 1175 • Fixed the document file name
- 1176 • Added section 4.5 “Enabling or Disabling the Printer Firmware Update Feature” and
1177 defined the “printer-firmware-update-enabled” attribute to allow a Client to have an
1178 easy way to detect support for the feature, and to provide the Printer and its
1179 Administrator with a way to disable the feature.

1180 **15.7 November 14, 2025**

1181 A number of updates following the review on 2025-10-30

- 1182 • Updated section 4 to create sub-sections
- 1183 • Updates sequence diagram in Figure 2
- 1184 • Updated 5.1 to correct missing attributes and to change the semantics to have a
1185 quick response and have the Client monitor the status via other attributes. This was
1186 one of the options in the slides and I chose this one and if we want to change it then
1187 we can change it. But we need to review section 4 in depth and that will help us nail
1188 down the semantics more quickly
- 1189 • Updates 5.2 as well
- 1190 • Removed the server-error-resource-unavailable status code, and gave server-error-
1191 firmware-update-in-progress 0x050D in its place.
- 1192 • Added several more “printer-state-reasons” to handle the Firmware Repository
1193 being unreachable and a Firmware update being unavailable, and also noted that at
1194 some phases of a Firmware update the IPP Printer will be entirely offline and unable
1195 to respond to Get-Printer-Attributes queries.

1196 15.8 October 29, 2025

- 1197
- Basically rewrote the entire thing again. Too many changes to name.

1198 15.9 May 6, 2025

1199 Updated with changes discussed at the November 2024 F2F and February 2025 F2F:

- 1200
- Refactored "printer-firmware-update" to eliminate the 1setOf collection and make all
- 1201 members top-level Printer Status attributes;
- 1202
- Rewrote the Check-For-Firmware-Update and Update-Printer-Firmware operations
- 1203 and some of the attribute descriptions; and
- 1204
- Updated the sequence diagram in section 4.

1205 15.10 October 24, 2024

1206 Updated with changes discussed in 2024-10-10 IPP WG teleconference and other offline

1207 discussions:

- 1208
- Refactored diagrams and descriptions of scenarios in section 4;
- 1209
- Created the new "printer-firmware-update" collection attribute, and moved many of
- 1210 the Printer Status attributes describing available new Firmware update under that so
- 1211 that a list of updates could be provided;

1212

 - Added the "printer-firmware-repository" attribute to allow the Firmware Repository to

1213 be configured; and

1214

 - Added the "printer-firmware-update-uri" operation attribute to allow a Client to

1215 specify a URI to a Firmware update file not on the configured Firmware Repository,

1216 and added a corresponding use case.

1217 15.11 October 10, 2024

1218 Updated with additional diagrams and several re-arrangements for section 4, which now

1219 includes both conceptual scenario description diagrams and sequence diagrams that show

1220 IPP operations where used.

1221 15.12 September 11, 2024

1222 Updated with feedback from the 2024-08-29 IPP Workgroup teleconference and other

1223 email feedback:

- 1224 • Updated the definition of "Firmware" using PWG 5110.1 definition and added
1225 reference;
- 1226 • Changed headings of use cases;
- 1227 • Moved Successful Firmware update from use cases to exceptions;
- 1228 • Updated the Model and updated the figures;
- 1229 • Added the "delay-update-until" and "delay-update-until-time" operation attributes,
1230 and changed the name of the operation to "Update-Printer-Firmware" operation;
- 1231 • Added the Check-For-Firmware-Update operation because having a Get-Printer-
1232 Attributes do the check as a side effect of requesting a particular Printer Status
1233 attribute seemed like a bad idea; and
- 1234 • Added additional keywords for the "printer-state-reasons" Printer Status attribute.

1235 **15.13 August 26, 2024**

1236 Initial revision.